



CHAPTER

15

Wave Motion

Sr. No.	Concept	Formulas	Other information
1.	Basic parameters of wave	Equation of travelling wave; $y = f(at \pm bx)$ $y = A \sin(\omega t - kx + \phi)$ Angular frequency, $\omega = 2\pi\nu = \frac{2\pi}{T}$ Angular wave number, $k = \frac{2\pi}{\lambda}$ Wave number, $\bar{\nu} = \frac{1}{\lambda} = \frac{k}{2\pi}$ Wave velocity, $v = \frac{\lambda}{T} = \frac{\lambda}{\frac{2\pi}{\omega}} = \frac{\omega}{2\pi} = \frac{\omega}{k}$	T = time period ν = frequency λ = wavelength A = amplitude
2.	Velocity of transverse waves in solids and strings	Velocity of transverse waves in a solid; $v = \sqrt{\frac{\eta}{\rho}}$ Velocity of transverse waves in a string; $v = \sqrt{\frac{T}{m}}$	η = modulus of rigidity ρ = density of solid m = mass per unit length of the string T = tension in the string
3.	Velocity of longitudinal waves	Velocity of longitudinal waves in a long rod; $v = \sqrt{\frac{Y}{\rho}}$ Velocity of longitudinal waves in liquid; $v = \sqrt{\frac{k}{\rho}}$	k = bulk modulus of elasticity ρ = density of the gas Y = Young's modulus

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4.	Velocity of sound in gases.	Newton's formula for the velocity of sound in a gas; $v = \sqrt{\frac{P}{\rho}}$ Laplace formula for the velocity of sound in a gas; $v = \sqrt{\frac{\gamma P}{\rho}}, \text{ where } \gamma = \frac{C_p}{C_v}$	P = pressure of the gas ρ = density of the gas γ = ratio of the specific heats
5.	Factors affecting the speed of sound in a gas	Effect of density, $v \propto \frac{1}{\sqrt{\rho}}$ or $\frac{v_1}{v_2} = \sqrt{\frac{\rho_2}{\rho_1}}$ Effect of temperature; $v = \sqrt{\frac{\gamma RT}{M}}$ $\therefore v \propto \sqrt{T} \text{ or } \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$	M = molar mass R = universal gas constant
6.	Equations of a plane progressive wave	Wave travelling along + x direction; (i) $y = A \sin (\omega t - kx), \quad k = \frac{2\pi}{\lambda}$ (ii) $y = A \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$ (iii) $y = A \sin \frac{2\pi}{\lambda} (vt - x)$ Wave travelling along – x direction; $y = A \sin (\omega t + kx)$ Or $y = A \sin 2\pi \left(\frac{t}{T} + \frac{x}{\lambda} \right) = A \sin \frac{2\pi}{\lambda} (vt + x)$	A = amplitude of the waves v = velocity of wave ω = angular frequency λ = wavelength of the wave x = displacement in time t T = time period of wave k = propagation constant
7.	Phase and phase difference of progressive wave	Thus phase, $\phi = 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$ The relation between phase difference ($\Delta\phi$) and path difference (Δx) is $\Delta\phi = \frac{-2\pi}{\lambda} \Delta x$	T = time period x = displacement in time t λ = wavelength

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8.	Instantaneous velocity and acceleration	Instantaneous particle velocity, $u = \frac{dy}{dt} = \frac{2\pi A}{T} \cos 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$ Maximum particle velocity or velocity amplitude, $u_0 = \frac{2\pi A}{T} = \omega A$ Instantaneous particle acceleration; $a = \frac{du}{dt} = -\frac{4\pi^2}{T^2} A \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) = -\omega^2 y$ Maximum particle acceleration or Acceleration amplitude; $a_0 = \frac{4\pi^2}{T^2} A = \omega^2 A$	A = amplitude of the wave λ = wavelength T = time period ω = angular frequency x = displacement
9	Relation between particle velocity and wave velocity	$u = -v \frac{dy}{dx}$ $\frac{dy}{dx}$ = slope of wave curve at given position and time	u = particle velocity at a given position at a given time v = wave velocity
10.	Principle of superposition of waves	Resultant displacement due to superposition; $y = y_1 + y_2 + y_3 + \dots + y_n$	$y_1, y_2, y_3, \dots, y_n$ = displacements of waves superposing each other at a point
11.	Interference for waves	Let the two waves are $y_1 = A_1 \sin(\omega t - kx_1)$ $y_2 = A_2 \sin(\omega t - kx_2)$ $A_{res} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \Delta\phi}$ We know that, $I \propto A^2$ $I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} \cos \Delta\phi$	A_1, A_2 = amplitude of the two waves ω = angular frequency k = wave number I = intensity of wave $\Delta\phi$ = phase difference between the waves

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12.	Constructive and destructive interference of waves	Constructive Interference $\Delta x = 0, \lambda, 2\lambda, 3\lambda, \dots n\lambda$ $n = 0, 1, 2, 3, \dots$ $\Delta\phi = 0, 2\pi, 4\pi, \dots 2n\pi$ $A_{\max} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos 0}$ $A_{\max} = A_1 + A_2$ $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$	Destructive Interference $\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots \frac{(2n-1)\lambda}{2}$ $n = 1, 2, 3, \dots$ $\Delta\phi = 0, 3\pi, 5\pi, \dots (2n-1)\pi$ $A_{\min} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \pi}$ $A_{\min} = A_1 - A_2$ $I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$	Δx = path difference $\Delta\phi$ = phase difference $A_{\max}$ = maximum amplitude $A_{\min}$ = minimum amplitude $I_{\max}$ = maximum intensity $I_{\min}$ = minimum intensity
		Ratio of the intensities; $\frac{I_{\max}}{I_{\min}} = \left(\frac{A_{\max}}{A_{\min}} \right)^2, \quad \frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2$		
13.	Equation of stationary/standing waves in string	String of length L fixed at ends $x = 0$ and $x = L$ Let, $y_1 = a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$ (incident wave) $y_2 = -a \sin 2\pi \left(\frac{t}{T} + \frac{x}{\lambda} \right)$ (reflected wave) Then stationary wave; $y = y_1 + y_2 = -2a \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi t}{T}$ Position of antinodes; $x = (2n+1) \frac{\lambda}{4}, n = 0, 1, 2, \dots$ Position of nodes; $x = \frac{n\lambda}{2}, n = 0, 1, 2, \dots$		λ = wavelength of the wave t = time x = displacement T = time period of wave
14.	Modes of vibrations of a string (fixed at both ends)	$L = \frac{n\lambda}{2}$ Fundamental frequency, $\nu = \frac{v}{2L} = \frac{1}{2L} \sqrt{\frac{T}{m}}$ Frequency of vibration in n^{th} mode; (When the stretched string vibrates in n loops) $\nu_n = \frac{n}{2L} \sqrt{\frac{T}{m}}$ For a string of diameter D and density ρ , $\nu = \frac{1}{LD} \sqrt{\frac{T}{\pi\rho}}$		ν = frequency L = length of the string T = tension in the string ρ = density of the material m = mass per unit length of the string

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15.	Vibration of string fixed at one end	Equation of standing wave; $y = 2a \sin kx \cdot \cos \omega t$ Frequency of the vibration: $f = \frac{(2n+1)}{4L} v_0, n = 0, 1, 2, \dots$ $f_0 = \frac{v_0}{4L}$ (first harmonic) $f_1 = \frac{3v_0}{4L}$ (Third harmonic and so on.....)	L = length of the string v_0 = velocity of the wave
16.	Vibration in open organ pipe	Fundamental frequency or frequency of first harmonic $f_1 = \frac{v}{2l}$ Frequency of first overtone or 3 rd harmonic $f_2 = 2 \frac{v}{2l} = 2f_1$ Frequency of second overtone or 5 th harmonic $f_3 = 3 \frac{v}{2l} = 3f_1$ Frequency in n^{th} normal mode of vibration: $\Rightarrow f_n = n f_1$ $\therefore f_1 : f_2 : f_3 : \dots = 1 : 2 : 3 : \dots$	v = velocity of sound l = length of the open pipe f_1, f_2, f_3 = frequencies of 1 st , 2 nd , 3 rd harmonic respectively.
17.	Vibration in closed organ pipe	Fundamental frequency or frequency of first harmonic, $f_1 = \frac{v}{4l}$ Frequency of first overtone or third harmonic, $f_2 = \frac{3v}{4l} = 3f_1$ Frequency of second overtone or fifth harmonic $f_3 = 5 \frac{v}{4l} = 5f_1$ Frequency in n^{th} normal mode of vibration: $\Rightarrow f_n = (2n-1)f_1$ $\therefore f_1 : f_2 : f_3 : \dots = 1 : 3 : 5 : \dots$	v = velocity of sound l = length of the closed pipe f_1, f_2, f_3 = frequencies in 1 st , 2 nd and 3 rd normal mode of vibration respectively.

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18.	Factors affecting frequency of the air column	(i) Length of air column, $n \propto \frac{1}{l}$ (ii) Temperature of air column, $n \propto \sqrt{T}$ (iii) Density, $n \propto \frac{1}{\sqrt{\rho}}$ (iv) Velocity of sound in air column, $n \propto v$	l = length of the air column T = temperature ρ = density v = velocity of sound
19.	Resonance tube & end correction	End correction; Thus, for a closed organ pipe replace L by $L + e$ $\therefore f_1 = \frac{v}{4(L + e)}$ And for an open organ pipe replace L by $L + 2e$ $\therefore f_2 = \frac{v}{2(L + 2e)}$ In closed organ pipe, length of air column of first resonance; $l_1 + e = \frac{\lambda}{4}$ At 2 nd resonance; $l_2 + e = \frac{3\lambda}{4}, \quad l_2 - l_1 = \frac{\lambda}{2}$ $e = \frac{l_2 - 3l_1}{4}$	l_1, l_2 = first and second resonance length λ = wave length of wave e = end correction
20.	Beats	Beat frequency = Number of beats per second = Difference in frequencies of two sources or $n = (n_1 - n_2)$ or $(n_2 - n_1)$	n = number of beats. n_1, n_2 = frequencies of different sound sources.